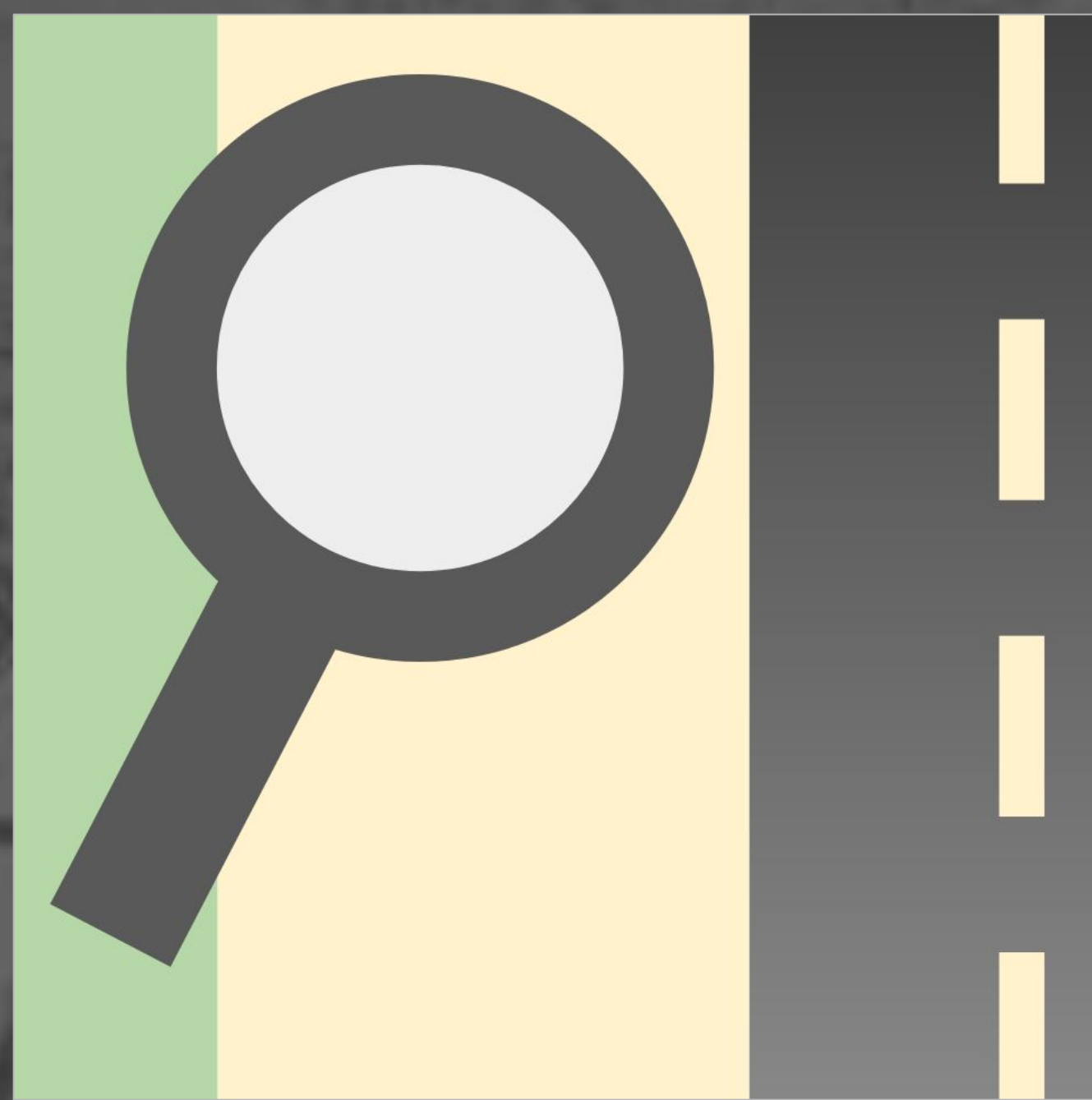




OpenSidewalkMap

a modular project, dedicated to portray OpenStreetMap *sidewalk networks*, its characteristics and applications

Sidewalks are essential for citizens worldwide; they mean safety and accessibility when in a good state. They are still under-represented at OSM. This work calls for the importance of their *detailed mapping*, which may provide public scrutiny and route optimization and be valuable for public administration

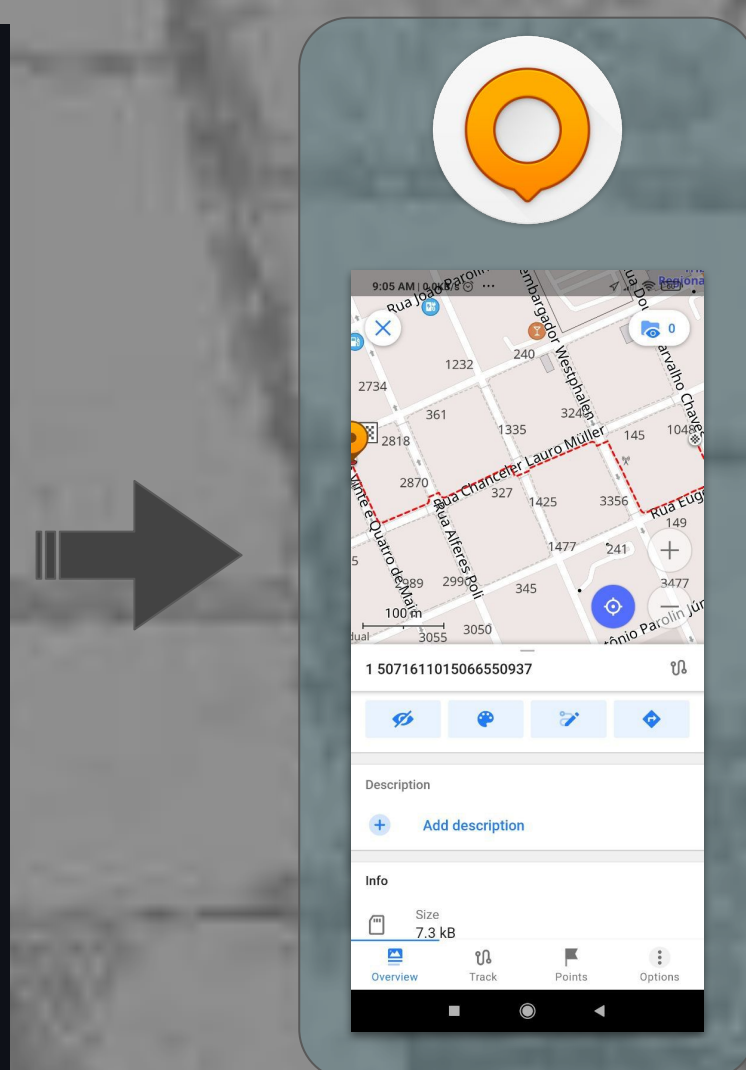
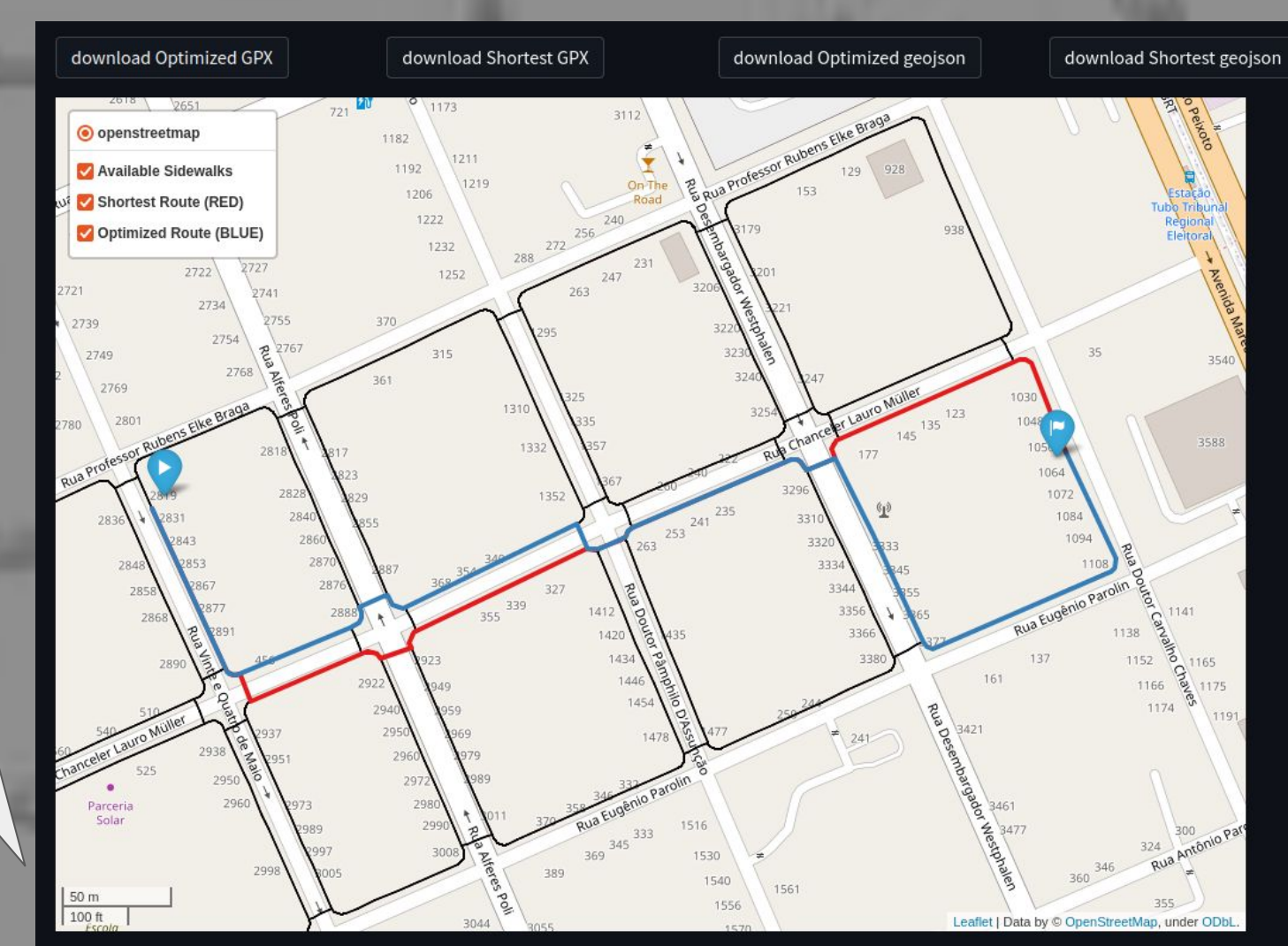
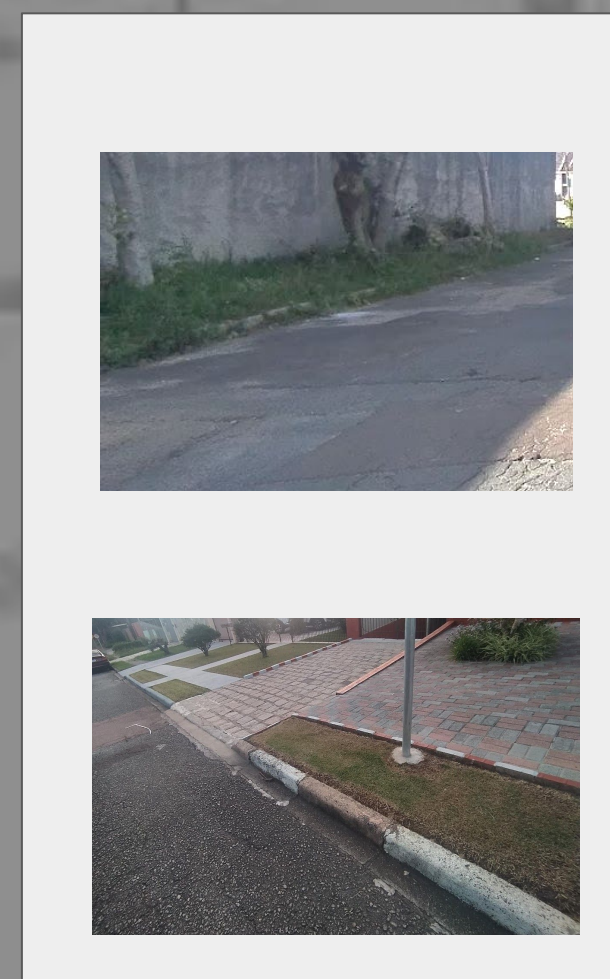
its modules are:

webmap

As the main module of the project, it presents an interactive cartographic representation of the three main subjects: **Sidewalks**; **Crossings**; and **Kerb Access Points**. It portrays the interest tags for accessibility, updating info and a direct link to each feature, enabling editing on iD. The catchphrase "*Sidewalk Mapping is a Cartographic Portrait of Structural Ableism*" is also motivated by the complete inadequacy of pavement on many Brazilian sidewalks.

optimized routing

Applications generally create "pedestrian" routes with some unrealistic traits: 1) traversing through left or right side as the same; 2) as if pedestrian would traverse street intersections; 3) only length and slope are considered. OSWM routing is a beta version that can trace on both sidewalk sides, also considering *surface*, *smoothness*, the type of *crossing* including if there are *ramps on kerbs*. Currently, it is optimized for wheelchair users, but in the future, it will include also a profile for visual impairment.



dashboard

Featuring interactive charts on the web version, it depicts some characteristics from OSWM interest data, primarily the number of occurrences and frequencies, and also presents the data's temporal accuracy. It also can be used to detect invalid tag values.

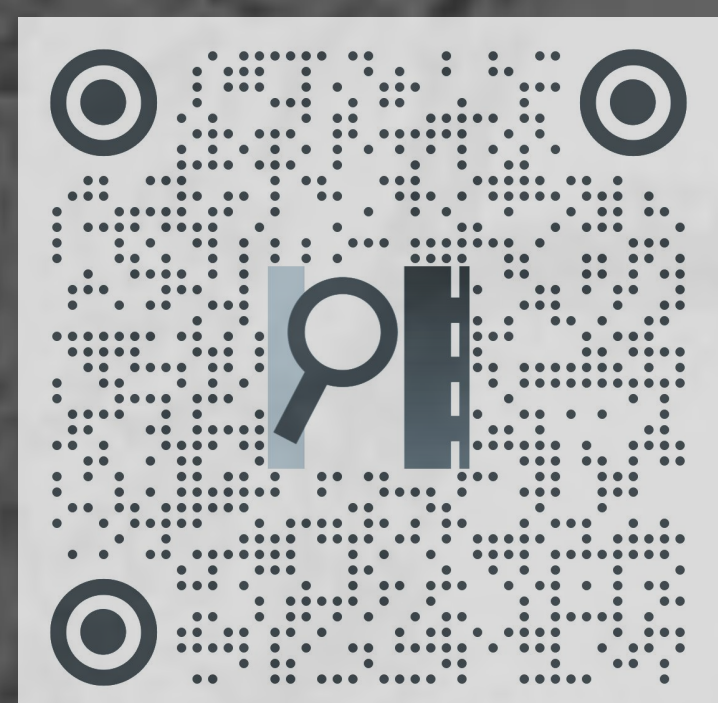
data quality tool

This Section is dedicated to find tagging errors in the features of interest in the Context of the project. Sometimes cases it is a clear mistake, but it can be just a misspelling or an uncommon value. In a future topological and geometric errors shall be included.

Category	Sidewalks	Crossings	Kerbs
improper_keys	63	7	99
uncanny_keys	58	2	9
keys_without_wiki	2	0	0
replaceable_values	0	307	0
wrong_misspelled_or_unlisted_values	36	14	13
invalid_characters	2	11	1

OSM ID (link)	key	value	commentary
883721682	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
883721685	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
883721687	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
883725829	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
883725831	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
883725869	kerb	normal	unlisted at accepted/known values, probably wrong/misspelled
883725870	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
8842953120	kerb	lowered-no	unlisted at accepted/known values, probably wrong/misspelled
8846409967	tactile_paving	bad	unlisted at accepted/known values, probably wrong/misspelled
8868954126	kerb	normal	unlisted at accepted/known values, probably wrong/misspelled
8890907326	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled
889106244	kerb	regular	unlisted at accepted/known values, probably wrong/misspelled

you can access the project at: kauevestena.github.io/opensidewalkmap_beta



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